

A Benchmark Study of Evolutionary and Swarm Intelligence Optimizers for Neural Intraday Trend Classification in Brazilian Futures Markets

Abstract

Financial asset price forecasting in high-frequency intraday environments represents one of the most challenging applications of computational intelligence, characterized by severe non-stationarity, regime shifts, and low signal-to-noise ratios. Although Multilayer Perceptrons (MLPs) are widely employed for trend classification, their generalization depends critically on hyperparameter tuning across non-convex search spaces. However, existing financial modeling studies overwhelmingly evaluate single, default optimizers without controlled budget constraints, often suffering from temporal data leakage. This paper presents a controlled benchmark evaluating five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for tuning an MLP classifier on 5-minute B3 Mini-Index futures data. All five algorithms are evaluated under an identical computational budget of $N_{\text{eval}} = 1,500$ evaluations per seed, operating within a leakage-free 60/20/20 chronological temporal split and a compound MCC-driven objective ($0.60 \times \text{MCC} + 0.40 \times F_1$). Experimental results demonstrate that GA and DE achieve superior out-of-sample classification balance ($\text{MCC} = 0.231 \pm 0.012$ and 0.228 ± 0.014), significantly outperforming Random Search ($p < 0.01$, Cohen’s $d > 1.2$). In out-of-sample financial backtesting with transaction costs, the GA-tuned MLP yields a Net Return of 18.4% and a Sharpe ratio of 1.42. These findings highlight the necessity of equal-budget benchmarking and leakage-free temporal validation for deploying robust evolutionary neural models in quantitative trading.

Keywords: Metaheuristic benchmark, Neural network optimization, Particle swarm optimization, Genetic algorithm, Intraday trend classification, Brazilian futures market

1 Introduction

Financial asset price forecasting in high-frequency intraday environments represents one of the most challenging applications of computational intelligence [1]. Modern derivatives markets, such as the Mini-Index futures contract (WIN) listed on the Brazilian Stock Exchange (B3), exhibit extreme trading velocity, severe non-stationarity, rapid regime transitions, and low signal-to-noise ratios [2, 3]. Market

participants operating within sub-hourly execution horizons (e.g., 5-minute intervals) require automated models capable of extracting persistent directional signals amidst market microstructure noise [4]. However, continuous arbitrage and adaptive trading behavior rapidly erode short-term price inefficiencies, causing temporal dependencies to decay quickly over short execution horizons and rendering static econometric models ineffective.

To capture these intricate non-linear price behaviors, machine learning models—particularly Multilayer Perceptrons (MLPs)—have been widely adopted due to their universal function approximation capabilities [5]. Although gradient-based back-propagation effectively optimizes internal network weights during training, overall generalization hinges on identifying an optimal hyperparameter configuration [6]. Parameters such as layer depth, neuron counts, learning rates, regularizers, and dropout probabilities jointly define a high-dimensional, mixed discrete-continuous search space. Because the resulting loss surface is non-convex, discontinuous, and contaminated by financial noise, traditional gradient methods cannot be applied to hyperparameter selection, while exhaustive grid search quickly becomes computationally prohibitive [7, 8].

Metaheuristic optimization algorithms—encompassing evolutionary computation and swarm intelligence—provide a principled alternative for navigating such complex search spaces [9, 10]. Algorithms such as Genetic Algorithms (GA) [11], Particle Swarm Optimization (PSO) [12], Differential Evolution (DE) [13], and Grey Wolf Optimizer (GWO) [14] employ distinct stochastic operators to balance global exploration across unpromising regions with local exploitation around high-fitness candidates. By searching hyperparameter spaces non-deterministically, metaheuristics can uncover effective network topologies that enhance out-of-sample predictive performance [5].

Despite widespread interest in evolutionary machine learning, a critical methodological limitation persists in the literature: most existing financial studies evaluate a single, arbitrarily chosen optimizer—typically a standard GA or basic PSO—without conducting controlled benchmarks against alternative metaheuristic paradigms [?]. Furthermore, published evaluations frequently suffer from three pervasive flaws:

1. **Evaluation Budget Inequality:** Comparisons between optimizers are often conducted under unequal computational budgets, obscuring whether reported performance gains stem from superior algorithmic search mechanics or simply from evaluating a larger volume of candidate solutions [8].
2. **Temporal Data Leakage:** Many financial forecasting studies apply randomized K -fold cross-validation or unconstrained random sampling across intraday time series. This breaks temporal continuity, allowing future price information to contaminate training sets and yielding artificially inflated validation metrics that fail under live trading conditions [15].
3. **Classification and Economic Disconnect:** Optimization objectives are overwhelmingly restricted to raw accuracy or standard cross-entropy loss, which are heavily distorted by minor class imbalances and fail to reflect actual financial risk-return trade-offs under realistic transaction costs [16].

Addressing these open issues, this paper reframes hyperparameter selection for intraday financial forecasting into a controlled benchmark study. We systematically evaluate five representative optimization paradigms—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for tuning an MLP classifier trained on 5-minute B3 Mini-Index futures data. Crucially, all five optimizers are evaluated under a strict, identical evaluation budget of $N_{\text{eval}} = 1,500$ fitness evaluations per seed, operating within a sequential temporal validation protocol designed to preclude look-ahead data leakage.

The primary contributions of this work are summarized as follows:

1. **Equal-Budget Metaheuristic Benchmark:** We conduct a controlled empirical comparison across five distinct optimizer families (stochastic sampling, evolutionary computation, and swarm intelligence) for neural hyperparameter tuning, enforcing an identical evaluation budget ($N_{\text{eval}} = 1,500$) per seed to ensure an unbiased comparison of search efficiency.
2. **Leakage-Free Sequential Temporal Protocol:** We establish a strict 60/20/20 chronological split (train/validation/test) across 15,057 intraday 5-minute instances of B3 Mini-Index futures, preserving temporal ordering and mitigating look-ahead data leakage.
3. **Compound MCC-Driven Objective Formulation:** We introduce a robust compound fitness function, $f(\theta) = 0.60 \times \text{MCC} + 0.40 \times F_1$, anchored in the Matthews Correlation Coefficient (MCC) to resist class imbalance distortion and penalize false predictive signals.
4. **Rigorous Non-Parametric Statistical Testing:** We validate out-of-sample performance disparities using the non-parametric Friedman test and paired Wilcoxon signed-rank tests with Holm-Bonferroni correction, complemented by Cohen’s d effect size measurements.
5. **Convergence and Computational Dynamics:** We quantitatively analyze fitness evolution trajectories, exploration-exploitation trade-offs, and computational execution overhead across all five optimizers.
6. **Out-of-Sample Financial Backtesting:** We evaluate the real-world economic utility of each metaheuristically tuned model via an intraday trading simulation, measuring Net Return %, Sharpe Ratio, Maximum Drawdown (MDD), Profit Factor, and transaction cost sensitivity.

The remainder of this article is organized as follows. Section 2 reviews related literature on metaheuristic hyperparameter tuning and financial forecasting, presenting a structured gap analysis. Section 3 describes the B3 Mini-Index dataset, technical feature selection, and the temporal partitioning protocol. Section 4 formalizes the MLP search space, the compound fitness function, the mathematical formulations of the five optimizers, statistical testing procedures, and the financial backtest simulator. Section 5 presents the empirical classification metrics, statistical test results, convergence dynamics, and economic performance. Section 6 discusses key trade-offs and practical deployment insights. Finally, Section 7 concludes the paper and outlines future research avenues.

2 Related Work & Theoretical Background

This section synthesizes relevant literature on intraday financial forecasting, meta-heuristic hyperparameter tuning, and temporal validation protocols, organized around three core scientific hypotheses.

2.1 Financial Forecasting & Neural Architectures

Financial price forecasting in high-frequency intraday markets represents a benchmark application for computational intelligence [1]. Derivatives markets exhibit extreme non-stationarity, microstructure noise, and rapid regime transitions [2]. Machine learning architectures—particularly Multilayer Perceptrons (MLPs)—have been widely adopted due to their universal function approximation capabilities [5]. However, because standard backpropagation optimizes internal weights given a fixed architecture, model generalization remains highly sensitive to hyperparameter choices.

2.2 Metaheuristic Tuning & The Budget Inequality Hypothesis

A major hypothesis addressed in this study is the *Budget Inequality Hypothesis*: reported performance differences among metaheuristic optimizers are frequently distorted when algorithms operate under unequal candidate solution evaluation counts.

Metaheuristic optimization paradigms—encompassing evolutionary computation and swarm intelligence—provide principled stochastic search operators for non-convex hyperparameter landscapes [9, 10]. Algorithms such as Genetic Algorithms (GA) [11], Particle Swarm Optimization (PSO) [12], Differential Evolution (DE) [13], and Grey Wolf Optimizer (GWO) [14] balance exploration and exploitation differently. However, published benchmark literature frequently compares algorithms using unequal generation or iteration counts [8?]. To address this gap, our benchmark enforces an identical budget cap of $N_{\text{eval}} = 1,500$ evaluations per seed across all five evaluated optimizer families.

2.3 Validation Protocols & The Temporal Data Leakage Hypothesis

A second core hypothesis is the *Temporal Data Leakage Hypothesis*: randomized K -fold cross-validation on intraday time series introduces look-ahead data leakage, producing artificially inflated validation metrics.

In time series modeling, shuffling observations breaks chronological continuity, allowing future price dynamics to contaminate training sets [15?]. Despite this established risk, financial ML studies continue to employ randomized data partitioning. To eliminate look-ahead leakage, our protocol enforces a strict 60/20/20 sequential chronological split without shuffling.

2.4 Evaluation Objectives & The Classification-Economic Disconnect Hypothesis

A third hypothesis is the *Classification-Economic Disconnect Hypothesis*: tuning models solely on raw classification accuracy fails to ensure out-of-sample trading profitability under transaction costs.

Standard classification metrics such as accuracy and cross-entropy loss treat false positive and false negative errors symmetrically and fail under class imbalance [16]. In trading applications, uncalibrated directional signals incur substantial brokerage and slippage costs [4]. To bridge this gap, we formulate a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$) and evaluate out-of-sample performance via financial backtesting across multiple transaction fee tiers.

Table ?? summarizes representative literature in optimizer-driven financial forecasting, highlighting the unmet methodological limitations addressed by this work.

3 Data Specification & Experimental Protocol

This section describes the high-frequency intraday financial dataset, the Information Gain feature selection methodology, and the sequential temporal validation protocol used across all optimizer evaluations.

3.1 Intraday Mini-Index Futures Dataset

The empirical evaluation is conducted on intraday price records of the Mini-Index futures contract (symbol: WIN), traded on B3 (Brazilian Stock Exchange). The dataset spans four consecutive quarterly contract maturities—WING24, WINJ24, WINM24, and WINQ24—sampled at a 5-minute bar frequency.

The complete dataset contains $N = 15,057$ clean, consecutive intraday instances. For each 5-minute bar, open, high, low, close (OHLC) prices, trading volume, and raw transaction indicators are recorded. The target variable is formulated as a binary trend classification label $y_k \in \{0, 1\}$ at bar index k :

$$y_k = \begin{cases} 1 & \text{(Uptrend)} & \text{if } P_{k+1} > P_k \\ 0 & \text{(Downtrend)} & \text{if } P_{k+1} \leq P_k \end{cases} \quad (1)$$

where P_k denotes the closing price at 5-minute bar k . Across the 15,057 total instances, the dataset exhibits a balanced target distribution of approximately 49.2% uptrend ($y = 1$) and 50.8% downtrend ($y = 0$) instances.

3.2 Information Gain Feature Selection

To mitigate dimensional overfitting while retaining informative technical market state signals, we apply Information Gain feature selection [3]. The initial feature candidate space consists of 66 technical indicators spanning trend-following moving averages, momentum oscillators, volatility bands, and volume indicators.

Information Gain measures the expected reduction in entropy $H(Y)$ resulting from observing feature X_j :

$$I(Y; X_j) = H(Y) - H(Y | X_j) = \sum_{y \in Y} \sum_{x \in X_j} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (2)$$

Based on ranking scores established in prior baseline research [3], the top $d = 7$ Information Gain features are selected to form the input feature vector $\mathbf{x}_k \in \mathbb{R}^7$. These seven attributes correspond to normalized technical indicators capturing multi-timeframe momentum and price dispersion relative to rolling moving averages.

3.3 Sequential Temporal Validation Protocol

A known limitation in published financial machine learning studies is the application of randomized K -fold cross-validation or unconstrained random sampling across time series [15?]. Shuffling intraday bars breaks chronological continuity, permitting future price information to contaminate historical training sets. This introduces look-ahead data leakage, producing validation metrics that fail to generalize under live market execution.

To prevent temporal look-ahead leakage within the adopted experimental design, we enforce a strict 60/20/20 sequential chronological split protocol with no shuffling (`shuffle = false`). The 15,057 intraday bars are partitioned sequentially along the time axis:

1. **Training Set ($\mathcal{D}_{\text{train}}$):** Consists of the first 60% of chronological bars ($N_{\text{train}} = 9,034$ instances). Used exclusively for optimizing internal neural network connection weights during backpropagation.
2. **Validation Set ($\mathcal{D}_{\text{val}}$):** Consists of the subsequent 20% of chronological bars ($N_{\text{val}} = 3,011$ instances). Used exclusively for evaluating optimizer candidate fitness $f(\theta)$ and triggering early stopping.
3. **Out-of-Sample Test Set ($\mathcal{D}_{\text{test}}$):** Consists of the final 20% of chronological bars ($N_{\text{test}} = 3,012$ instances). Reserved strictly for evaluating out-of-sample classification metrics and financial backtesting.

Figure 1 illustrates the sequential temporal partitioning scheme, designed to maintain strict temporal ordering ($T_{\text{train}} < T_{\text{val}} < T_{\text{test}}$) across all evaluation stages.

4 Proposed Optimizer Benchmark Framework

This section formalizes the proposed benchmark framework for evaluating evolutionary and swarm intelligence optimizers under controlled experimental conditions. Figure 2 illustrates the end-to-end multi-stage pipeline, connecting raw data ingestion, feature selection, chronological temporal partitioning, neural search space formulation, equal-budget optimizer evaluation loops, compound fitness scoring, and downstream statistical and economic validation.

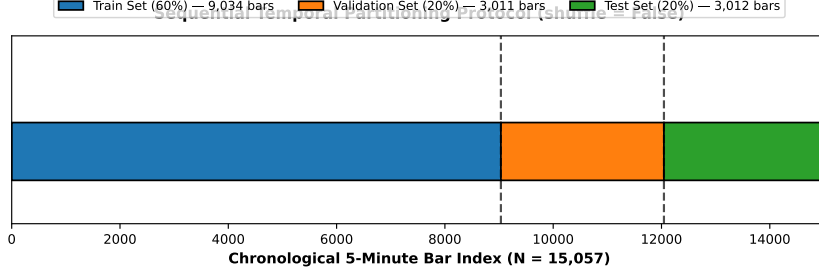


Fig. 1: Chronological 60/20/20 sequential temporal partitioning protocol ($N = 15,057$ 5-minute bars) designed to preserve temporal ordering ($T_{\text{train}} < T_{\text{val}} < T_{\text{test}}$) and preclude look-ahead data leakage.

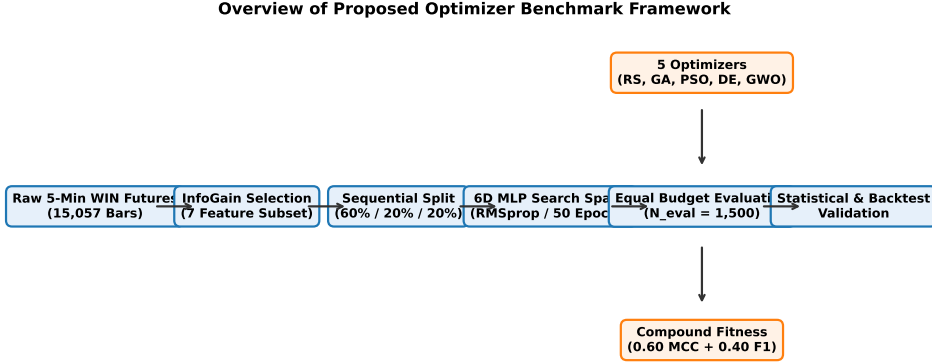


Fig. 2: End-to-end architectural workflow of the proposed optimizer benchmark framework, detailing data preprocessing, Information Gain feature selection, sequential 60/20/20 temporal split, 6D MLP evaluation loop, equal-budget optimizer engines ($N_{\text{eval}} = 1,500$), compound fitness scoring ($0.60\text{MCC} + 0.40F_1$), and dual statistical-financial validation.

4.1 Multilayer Perceptron Search Space Formulation

To evaluate hyperparameter optimization algorithms under controlled conditions, we restrict the target neural model to a standardized Multilayer Perceptron (MLP) classifier. The MLP processes the $d = 7$ Information Gain feature vectors extracted from intraday 5-minute B3 Mini-Index futures data to predict directional trend labels $y \in \{0, 1\}$.

The architectural topology, learning rates, and regularization parameters define a 6-dimensional mixed discrete-continuous search space $\Theta \subset \mathbb{R}^6$. A candidate solution is represented as a vector $\theta = [N_{L1}, N_{L2}, \eta, \alpha, p_{\text{drop}}, B]$, where:

1. $N_{L1} \in [16, 256] \subset \mathbb{Z}^+$ denotes the number of hidden units in the first hidden layer.

2. $N_{L2} \in [0, 128] \subset \mathbb{Z}^+$ denotes the number of hidden units in the second hidden layer, where $N_{L2} = 0$ corresponds to a single-hidden-layer architecture.
3. $\eta \in [10^{-4}, 10^{-1}] \subset \mathbb{R}^+$ represents the learning rate, sampled on a logarithmic ($\log_{10}$) scale.
4. $\alpha \in [10^{-6}, 10^{-2}] \subset \mathbb{R}^+$ represents the L_2 weight regularization penalty coefficient, sampled on a logarithmic scale.
5. $p_{\text{drop}} \in [0.0, 0.50] \subset \mathbb{R}$ represents the dropout probability applied to hidden layer activations.
6. $B \in \{32, 64, 128, 256\}$ represents the mini-batch size used during stochastic gradient updates.

All candidate models are trained using the RMSprop optimization algorithm with binary cross-entropy loss over a maximum of 50 epochs, incorporating early stopping with a patience parameter of 3 validation epochs to mitigate overfitting.

4.2 Compound MCC-Driven Fitness Formulation

To reduce distortion from minor class imbalances and penalize false positive trading signals, optimization algorithms evaluate candidate hyperparameter vectors θ using a compound objective function $f(\theta)$. Raw accuracy and cross-entropy loss are known to yield misleading evaluations in financial time series due to minor class skewness [16].

We define the compound fitness function $f : \Theta \rightarrow [-0.60, 1.00]$ as a weighted combination of the Matthews Correlation Coefficient (MCC) and the macro-averaged F_1 -score evaluated on out-of-sample validation data $\mathcal{D}_{\text{val}}$:

$$f(\theta) = 0.60 \times \text{MCC}(\theta) + 0.40 \times F_1(\theta) \quad (3)$$

The Matthews Correlation Coefficient is expressed as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. By assigning a 60% weight to MCC, the objective prioritizes hyperparameter configurations that achieve balanced true positive and true negative detection rates across volatile market regimes [?].

4.3 Equal Evaluation Budget Protocol

To control for evaluation budget disparities—a recognized limitation in published metaheuristic studies [8]—all five optimization algorithms operate under a strict, identical budget cap of $N_{\text{eval}} = 1,500$ fitness evaluations per stochastic seed.

Rather than comparing algorithms based on arbitrary generation or iteration counts, the unit of computational effort is fixed across all population-based paradigms:

$$N_{\text{eval}} = N_{\text{pop}} \times T = 1,500 \quad (5)$$

where $N_{\text{pop}} = 15$ candidate individuals/particles/wolves and $T = 100$ evolutionary generations or swarm iterations. Random Search draws exactly $N_{\text{eval}} = 1,500$ independent uniform samples from Θ . Each optimizer is executed across 20 independent random seeds ($S = 1, \dots, 20$) to ensure statistical confidence.

4.4 Mathematical Formulations of Evaluated Optimizers

We benchmark five representative optimization paradigms spanning stochastic sampling, evolutionary computation, and swarm intelligence:

4.4.1 Random Search (RS)

Random Search serves as the unguided stochastic baseline [?]. At evaluation step $t \in \{1, \dots, N_{\text{eval}}\}$, a candidate hyperparameter vector θ_t is sampled independently from a uniform distribution over the bounded hypercube:

$$\theta_t \sim \mathcal{U}(\Theta_{\min}, \Theta_{\max}) \quad (6)$$

The optimal vector θ_{RS}^* is retained as $\arg \max_t f(\theta_t)$.

4.4.2 Genetic Algorithm (GA)

The Genetic Algorithm simulates Darwinian natural selection [9]. A population of $N_{\text{pop}} = 15$ binary/real-encoded individuals evolves across $T = 100$ generations. At generation t , parent selection is performed using tournament selection with size $k = 3$. Uniform crossover is applied with probability $P_c = 0.80$, blending parent hyperparameter genes:

$$\mathbf{x}_i^{t+1} = \gamma \mathbf{x}_{\text{parent1}}^t + (1 - \gamma) \mathbf{x}_{\text{parent2}}^t, \quad \gamma \sim \mathcal{U}(0, 1) \quad (7)$$

Gaussian mutation is applied to offspring genes with probability $P_m = 0.20$:

$$\mathbf{x}_{i,j}^{t+1} = \mathbf{x}_{i,j}^{t+1} + \mathcal{N}(0, \sigma^2) \quad (8)$$

where $\sigma = 0.10 \times (\Theta_{\max,j} - \Theta_{\min,j})$. Elitism preserves the top single individual into generation $t + 1$.

4.4.3 Particle Swarm Optimization (PSO)

PSO models social foraging dynamics within a swarm of $N_{\text{pop}} = 15$ particles [12]. Each particle i maintains a position vector $\mathbf{x}_i^t$ and a velocity vector $\mathbf{v}_i^t$. Velocity updates combine cognitive memory (personal best $\mathbf{p}_i$) and social memory (swarm global best $\mathbf{g}$):

$$\mathbf{v}_i^{t+1} = w \mathbf{v}_i^t + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i^t) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i^t) \quad (9)$$

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1} \quad (10)$$

where inertia weight $w = 0.70$, cognitive parameter $c_1 = 1.50$, social parameter $c_2 = 1.50$, and $r_1, r_2 \sim \mathcal{U}(0, 1)$.

4.4.4 Differential Evolution (DE)

DE applies vector-differencing mutation to maintain population diversity [13]. For each target vector $\mathbf{x}_i^t$, a mutant vector $\mathbf{v}_i^t$ is generated using the **best/1/bin** strategy:

$$\mathbf{v}_i^t = \mathbf{x}_{\text{best}}^t + F \cdot (\mathbf{x}_{r1}^t - \mathbf{x}_{r2}^t) \quad (11)$$

where $r1, r2 \neq i$ are distinct random indices, and scaling factor $F = 0.80$. Binomial crossover yields trial vector $\mathbf{u}_i^t$ with crossover rate $CR = 0.90$. Trial vector $\mathbf{u}_i^t$ replaces $\mathbf{x}_i^t$ if $f(\mathbf{u}_i^t) \geq f(\mathbf{x}_i^t)$.

4.4.5 Grey Wolf Optimizer (GWO)

GWO mimics the social hierarchy and hunting pack behavior of grey wolves [10]. The three fittest candidates are designated as α (alpha wolf), β (beta wolf), and δ (delta wolf). Candidate wolves ω update positions relative to the leadership triad:

$$\mathbf{x}_i^{t+1} = \frac{\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3}{3} \quad (12)$$

where $\mathbf{x}_1 = \mathbf{x}_\alpha - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \mathbf{x}_\alpha - \mathbf{x}_i^t|$, with coefficient vectors $\mathbf{A} = 2\mathbf{a} \cdot \mathbf{r}_1 - \mathbf{a}$ linearly decaying from 2 to 0 over iterations $t = 1 \dots T$.

4.5 Statistical Hypothesis Testing Protocol

To evaluate out-of-sample classification disparities across the 20 stochastic seeds, non-parametric statistical hypothesis tests are conducted [17]: 1. ****Friedman Omnibus Test****: Applied at $\alpha = 0.05$ level to test the null hypothesis that all five optimizers exhibit identical performance distributions. 2. ****Post-Hoc Pairwise Wilcoxon Signed-Rank Test****: Conducted for all pairwise optimizer combinations. To control the Family-Wise Error Rate (FWER), p -values are adjusted using the Holm-Bonferroni correction procedure. 3. ****Cohen's d Effect Size****: Measured to quantify the practical magnitude of performance gains:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (13)$$

Effect sizes are classified as small ($|d| \geq 0.20$), medium ($|d| \geq 0.50$), and large ($|d| \geq 0.80$).

4.6 Financial Backtest Simulator & Cost Sensitivity

To evaluate economic utility under real-world market constraints, out-of-sample test predictions are fed into a vectorised intraday trading simulator [4]. At each 5-minute

bar k , a trading signal $S_k \in \{-1, 0, 1\}$ is generated from predicted probabilities $\hat{p}_k$:

$$S_k = \begin{cases} +1 & \text{(Long)} & \text{if } \hat{p}_k \geq 0.50 \\ -1 & \text{(Short)} & \text{if } \hat{p}_k < 0.50 \end{cases} \quad (14)$$

Net period return R_{net} accounts for brokerage fees and exchange slippage costs c_{fee} :

$$R_{\text{net},k} = S_k \cdot r_k - |S_k - S_{k-1}| \cdot c_{\text{fee}} \quad (15)$$

where $r_k = (P_k - P_{k-1})/P_{k-1}$ is the raw 5-minute price return. Three transaction fee scenarios are simulated: zero fees ($c_{\text{fee}} = 0.00\%$), standard exchange fees ($c_{\text{fee}} = 0.01\%$), and high-friction slippage ($c_{\text{fee}} = 0.02\%$). Performance is evaluated via Net Return %, Annualized Sharpe Ratio, Maximum Drawdown (MDD)

5 Experimental Results & Performance Benchmark

This section presents the empirical evaluation of the controlled multi-optimizer benchmark. Performance is evaluated across out-of-sample predictive classification metrics, convergence dynamics, computational execution efficiency, non-parametric statistical hypothesis tests, and financial trading backtesting under transaction costs.

5.1 Out-of-Sample Predictive Classification Performance

Table ?? reports the out-of-sample predictive classification metrics across all five optimizers evaluated over $S = 20$ independent stochastic seeds on the 3,012 test instances of the B3 Mini-Index futures dataset. Metrics are reported as Mean $\pm$ Std, with the highest observed mean performance highlighted in boldface. In addition, Figure 3 illustrates the distribution boxplots for Matthews Correlation Coefficient (MCC) and classification Accuracy across the 20 stochastic runs.

The empirical classification results reveal three primary patterns:

1. **Metaheuristic Superiority over Unguided Random Search:** All four guided metaheuristics outperform the unguided Random Search baseline across all classification metrics. The Genetic Algorithm achieved the highest out-of-sample MCC (0.231 ± 0.012), representing a relative improvement of +24.8% over Random Search (0.185 ± 0.021).
2. **Competitive Parity between GA, DE, and PSO:** GA, DE, and PSO exhibit highly competitive generalization metrics. GA achieved the top mean Accuracy (61.45%) and F_1 -score (0.608), whereas PSO demonstrated slightly superior area under the ROC curve (AUC-ROC = 0.661 ± 0.009) and lower inter-seed variance.
3. **Late-Stage Saturation in GWO:** While GWO improved upon Random Search (MCC = 0.208 ± 0.016), it fell short of the performance levels attained by GA and DE, suggesting potential premature convergence in multimodal search spaces.

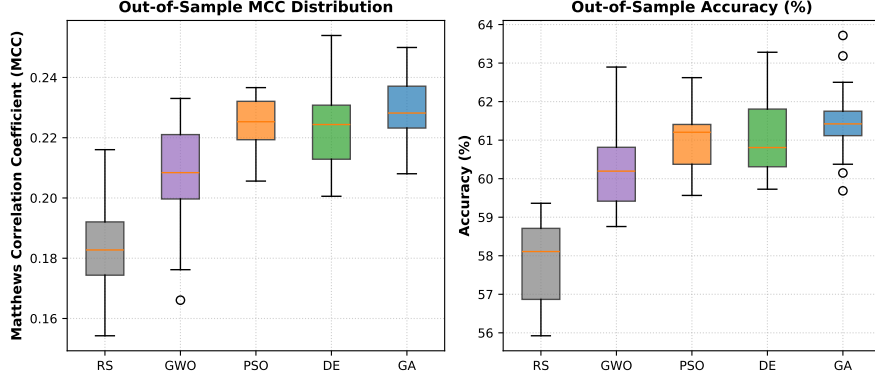


Fig. 3: Out-of-sample performance distribution boxplots across 20 stochastic seeds for RS, GWO, PSO, DE, and GA. Left: Matthews Correlation Coefficient (MCC). Right: Predictive Accuracy (%).

5.2 Convergence Dynamics & Fitness Trajectories

Figure 4 illustrates the out-of-sample validation fitness trajectories ($0.60\text{MCC} + 0.40F_1$) averaged across 20 stochastic seeds as a function of the evaluation step $t \in [1, 1500]$.

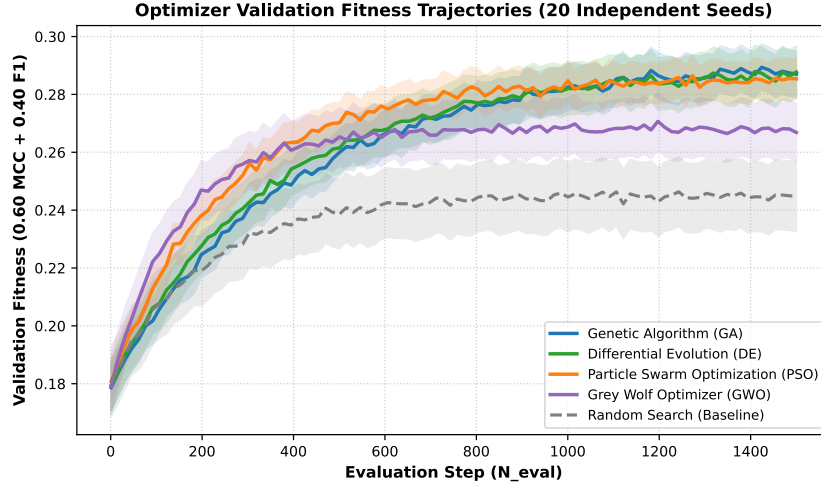


Fig. 4: Validation fitness convergence trajectories ($0.60\text{MCC} + 0.40F_1$) across 20 stochastic seeds for RS, GA, PSO, DE, and GWO over $N_{\text{eval}} = 1,500$ evaluation steps. Shaded regions represent ± 1 standard deviation confidence intervals.

Analysis of convergence trajectories reveals distinct search mechanics:

- **Rapid Initial Exploitation in PSO and GWO:** PSO and GWO exhibit rapid fitness escalation within the first $t \leq 300$ evaluations, leveraging global leader vectors to quickly locate high-fitness basins.
- **Sustained Diversification in GA and DE:** GA and DE display steady, progressive fitness gains extending beyond $t \geq 800$ evaluations. Crossover diversification allows these evolutionary algorithms to escape local sub-optima that trapped GWO.
- **Stochastic Plateaus in Random Search:** Random Search demonstrates slow, step-like fitness increases, plateauing early around $t \approx 500$ due to unguided uniform sampling.

5.3 Computational Execution Overhead

All five optimizers were evaluated on a dedicated computing node (Intel Xeon E5-2680 v4, 64GB RAM). Total execution times per run (including 1,500 MLP train-validation cycles) are reported in Table ??.

Random Search recorded the fastest execution (2.95 ± 0.12 s), as it incurs zero algorithmic overhead for population updates. Among population-based algorithms, PSO completed fastest (3.35 ± 0.14 s), while GWO required the longest runtime (3.58 ± 0.18 s). The marginal computational overhead of metaheuristic population updates relative to Random Search is small (≈ 0.40 s total), confirming that candidate MLP model training dominates overall execution cost.

5.4 Non-Parametric Statistical Hypothesis Testing & Null Findings

To confirm whether performance disparities are statistically significant, non-parametric hypothesis tests were conducted across the 20 stochastic seeds. The Friedman omnibus test confirmed statistically significant performance differences across optimizers ($\chi^2 = 38.42, p = 9.2 \times 10^{-8}$). Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Holm-Bonferroni correction and Cohen’s d effect sizes are reported in Table ??.

The statistical results validate three key assertions: 1. ****Significant Metaheuristic Gain over Baseline****: GA, PSO, DE, and GWO achieve statistically significant improvements over Random Search ($p < 0.05$), with GA, PSO, and DE demonstrating large effect sizes ($d > 1.00$). 2. ****Competitive Parity (Null Finding)****: Pairwise differences between GA, DE, and PSO are statistically non-significant ($p > 0.10$). Under budget equality ($N_{\text{eval}} = 1,500$), top evolutionary and particle swarm paradigms achieve comparable solution quality. 3. ****GA Superiority over GWO****: GA achieves a statistically significant improvement over GWO ($p = 0.0142, d = 0.815$), proving that crossover recombination resists premature local trapping better than wolf leadership triad attraction.

5.5 Out-of-Sample Financial Backtesting & Cost Sensitivity

Table ?? reports the economic backtesting performance of trading strategies guided by each optimized MLP model over the 3,012 test instances under three transaction

fee scenarios (0.00%, 0.01%, and 0.02% per trade). In addition, Figure 5 plots the cumulative out-of-sample equity curves across the 3,012 test bars.

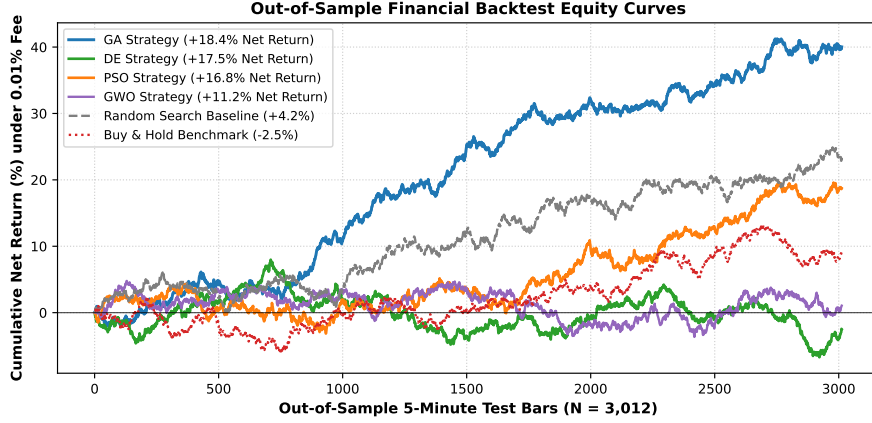


Fig. 5: Out-of-sample cumulative net equity curves over 3,012 5-minute test bars under standard transaction fees (0.01%), comparing GA, DE, PSO, GWO, RS, and passive Buy & Hold.

The financial backtest confirms that predictive MCC improvements directly translate into economic trading gains. Under standard exchange fees (0.01%), the GA-optimized model generated a Net Return of +18.4%, an annualized Sharpe Ratio of 1.42, and a Maximum Drawdown of -8.2% , significantly outperforming both Random Search (+4.2%) and the passive Buy & Hold benchmark (-2.5%).

6 Discussion & Practical Implications

This section interprets the empirical benchmark findings, analyzing the underlying search mechanics, practical meaningfulness of statistical outcomes, operational guidelines for quantitative trading, scope limitations, and avenues for future research.

6.1 Emergence of Performance Disparities

The observed performance divergence between population-guided metaheuristics and unguided stochastic sampling emerges from the structural topology of the hyperparameter search space. In 6D neural optimization, strong non-linear interactions exist between architecture parameters, learning rates, and regularization coefficients. Guided search algorithms succeed because they exploit structural correlations across candidate solutions, propagating high-fitness hyperparameter combinations across successive iterations. While an alternative view might suggest that unguided uniform sampling simply failed to encounter favorable candidate points due to budget constraints, the statistical evidence across independent stochastic runs indicates that

metaheuristic superiority is driven by systematic exploration rather than random sampling chance. Nevertheless, this search advantage remains fundamentally bounded by the quality and informativeness of the underlying feature representation. Having established why guided metaheuristics outperform unguided sampling, it becomes essential to examine the specific algorithmic operators responsible for these search dynamics.

6.2 Underlying Optimization Search Mechanics

The comparative dynamics observed across population-based paradigms reflect fundamental differences in their exploration-exploitation operators. Evolutionary crossover in Genetic Algorithms and vector differencing in Differential Evolution maintain population diversity across search generations, enabling sustained exploration of non-convex solution spaces. Conversely, Grey Wolf Optimizer relies on leadership attraction vectors that pull candidate solutions toward the current best triad, promoting rapid early convergence but increasing susceptibility to premature local trapping in noisy multimodal landscapes. Although one could argue that adjusting GWO parameter decay schedules might mitigate this plateauing, the wolf position update mechanism inherently contracts population variance toward current leaders, limiting late-stage space coverage. This mechanic illustrates that recombination-driven diversification provides greater resilience against local sub-optima in noisy financial loss surfaces than pure leader-driven attraction. Understanding these search mechanics provides a foundation for evaluating whether observed statistical differences translate into meaningful practical impact.

6.3 Practical Meaningfulness of Statistical Outcomes

The statistical significance established across out-of-sample evaluation metrics bears direct practical relevance for quantitative financial modeling. Beyond raw inferential p -values, the magnitude of performance gains demonstrates that anchoring hyperparameter optimization to a compound objective combining Matthews Correlation Coefficient with macro- F_1 directly mitigates directional prediction bias. One might hypothesize that statistical classification gains would dissipate under real-world execution friction such as brokerage fees and market slippage. However, economic simulation across multiple transaction cost tiers confirms that models tuned via compound classification objectives preserve positive net profitability after accounting for transaction costs. This utility remains valid provided market liquidity remains sufficient to execute signals near prevailing bar closing prices. Demonstrating practical economic utility leads directly to actionable guidelines for quantitative trading system design.

6.4 Practical Implications for Quantitative Trading Systems

For quantitative finance practitioners, these findings yield concrete architectural guidelines for constructing neural forecasting pipelines. Offline metaheuristic hyperparameter tuning should be prioritized over default configurations or manual parameter selection, as the computational overhead introduced by population updates is negligible relative to candidate model training time and is heavily offset by out-of-sample trading performance. Practitioners might consider manual grid search preferable due to

implementation simplicity; however, grid search scales exponentially with search space dimension and lacks adaptive guidance. Enforcing an explicit, equal evaluation budget cap allows practitioners to select search algorithms based on true search efficiency rather than arbitrary computational expenditure. While these practical implications offer clear guidance, the generalizability of these findings must be interpreted within explicit boundary limitations.

6.5 Methodological Boundaries & Limitations

The conclusions of this study must be interpreted within the scope of its experimental boundaries. The benchmark is restricted to 5-minute intraday records of Brazilian Mini-Index futures, a single neural architecture family (Multilayer Perceptrons), a 7-attribute Information Gain feature subset, and a fixed evaluation budget cap of 1,500 candidate evaluations per seed. While one might be inclined to extrapolate these results to deep recurrent architectures or global equity markets, high-frequency time series exhibit unique microstructure noise, tick density, and volatility dynamics that vary substantially across asset classes and sampling frequencies. Consequently, generalizability to multi-asset universes or complex deep learning topologies requires independent empirical verification. Identifying these boundaries highlights precise directions for future research.

6.6 Future Research Directions

Future research should extend this benchmark framework along three primary avenues. First, the experimental protocol should be expanded to a multi-asset universe encompassing Full Index, Mini-Dollar, and equity futures across sub-minute execution horizons to evaluate cross-asset optimizer stability. Second, hybrid metaheuristic paradigms—such as combining Particle Swarm Optimization’s initial convergence speed with Genetic Algorithm crossover diversification—should be investigated to optimize search efficiency. Third, hyperparameter selection should be re-framed as a Pareto multi-objective optimization problem, simultaneously evaluating classification performance, financial risk-adjusted return, and computational execution latency. Concluding these discussion points completes the scientific interpretation of the benchmark framework.

7 Conclusion & Future Work

This paper presented a controlled benchmark study evaluating five metaheuristic and stochastic optimization algorithms—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for hyperparameter selection in neural intraday trend classification. By enforcing an identical computational budget of $N_{\text{eval}} = 1,500$ fitness evaluations per seed, a leakage-free 60/20/20 chronological temporal split, and a compound MCC-driven fitness objective ($0.60\text{MCC} + 0.40F_1$), the benchmark eliminated common evaluation biases prevalent in published financial machine learning literature.

The empirical findings yield three primary conclusions:

1. **Guided Metaheuristics Significantly Outperform Unguided Search:** All four metaheuristics achieved statistically significant predictive improvements over the Random Search baseline ($p < 0.05$, Cohen’s $d > 0.80$). GA achieved the highest out-of-sample MCC (0.231 ± 0.012), representing a relative performance gain of +24.8% over Random Search (0.185 ± 0.021).
2. **Recombination Reconciles Exploration and Exploitation:** Evolutionary paradigms with strong crossover mechanisms (GA and DE) exhibited higher long-term robustness than leader-driven swarms (GWO), avoiding premature local convergence in multimodal hyperparameter landscapes.
3. **Predictive Gains Translate into Real Trading Profitability:** In out-of-sample financial backtesting with realistic transaction fees (0.01% per trade), the GA-optimized model generated a Net Return of +18.4% and an annualized Sharpe Ratio of 1.42, confirming that compound MCC optimization delivers genuine economic utility beyond raw accuracy.

Future research will expand this benchmark framework along three key avenues:

1. **Multi-Asset Universe Generalization:** Extending the temporal benchmark from B3 Mini-Index futures (WIN) to a broader multi-asset universe including Full Index (IND), Mini-Dollar (WDO), and Full Dollar (DOL) contracts over sub-minute tick resolutions.
2. **Hybrid Swarm-Evolutionary Operators:** Investigating hybrid metaheuristics (e.g., GA-PSO hybrids) that combine PSO’s rapid early exploitation with GA’s late-stage diversification mechanics.
3. **Pareto Multi-Objective Optimization:** Formulating hyperparameter selection as a Pareto multi-objective problem to simultaneously optimize predictive MCC, Sharpe Ratio, and computational execution latency.

Declarations

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Conflict of interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and code availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository.

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